Ku Kang
CBRN Defense Research Institute, Republic of Korea
bisu9082@gmail.com

The Editors, *Environmental Science & Technology*

Dear Editors,

We submit the Article "Can Structure-Based Models Clear Untested PFAS? External Validation Reveals Endpoint-, Representation-, and Distribution-Dependent Reliability" for consideration in *Environmental Science & Technology*.

The manuscript addresses a regulatory decision that is often left implicit in machine- learning screening: whether a model that prioritizes chemicals can also justify setting untested PFAS aside as low concern. We distinguish broad-space cross-validation from external PFAS validation and evaluate both point prediction and interval coverage. After removing PFAS matches, duplicate desalted structures, invalid inventory structures, and measured-inventory overlap, the analysis is based on 3,788 unique non-PFAS training structures, 138 measured PFAS, and 18,851 unique untested inventory structures.

The central result is cautionary but actionable. Broad-space performance is moderate ($R^2 = 0.39 \pm 0.02$), yet external PFAS prediction fails ($R^2 = -0.58$; $\rho = -0.07$). At nominal 90% coverage, broad marginal intervals cover only 0.659 of PFAS on average. PFAS-conditional split calibration restores held-out coverage to 0.922, but the resulting intervals are too wide to clear any in-domain untested structure. We explicitly qualify the deployment statement by the measured-to-untested exchangeability assumption. We also show that endpoint and representation choices materially alter PFAS transfer, so reliability belongs to the complete modeling pipeline rather than to the algorithm alone.

We believe the work fits ES&T because it couples environmental data curation, aquatic hazard prediction, uncertainty validation, and decision consequences for a high-priority chemical class. The revised package contains structure-locked data, repeat-level results, and scripts that generate every reported number and figure. The manuscript is original, is not under consideration elsewhere, and is approved by all authors. The authors declare no competing financial interest.

Sincerely,
Ku Kang, on behalf of all authors